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TUM School of Natural Sciences
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SONAR: A Physics-constrained Neural Representation for X-ray Dark-field CT

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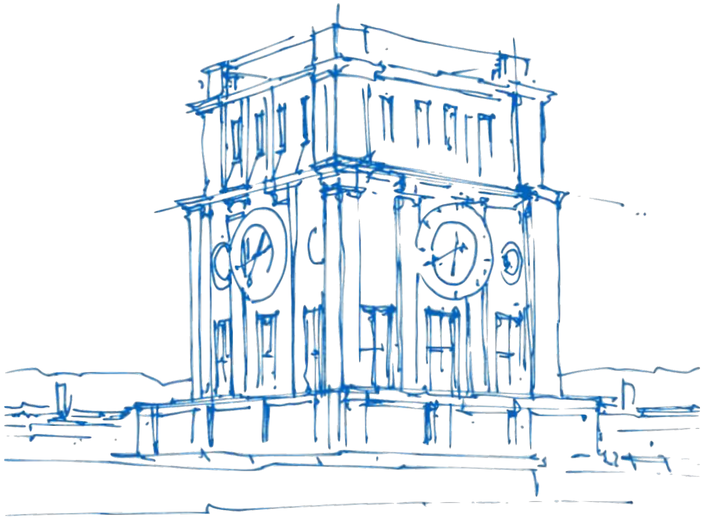
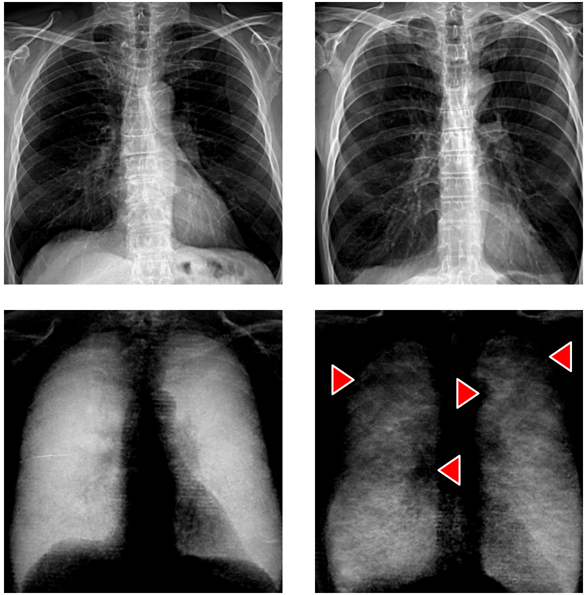


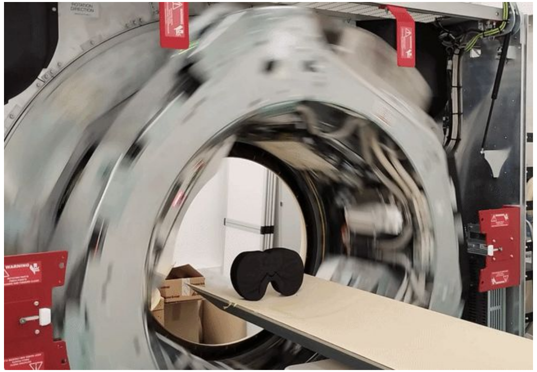
Uhrenturm der TUM

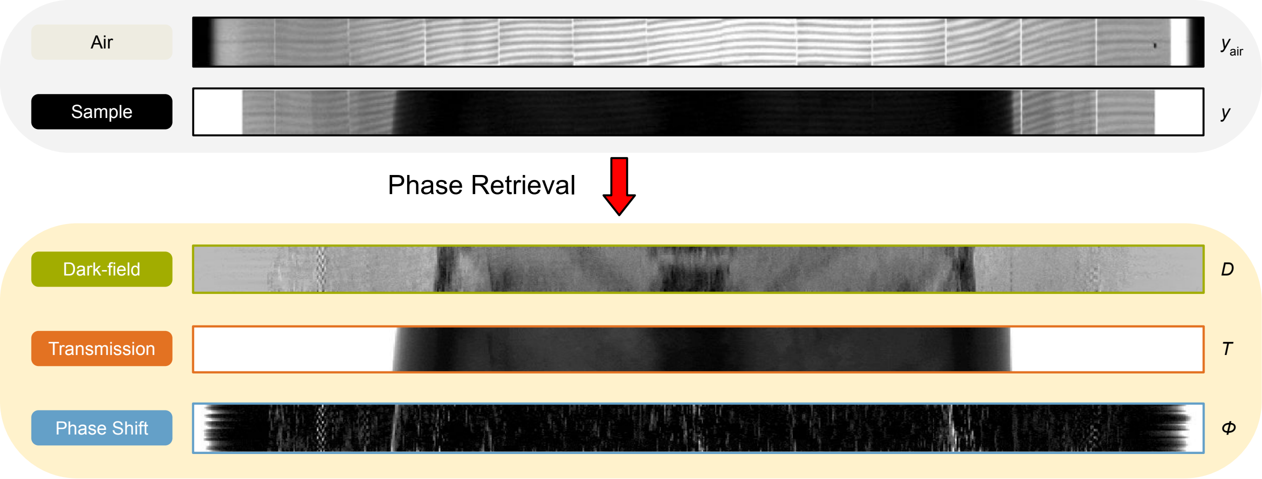
Dark-field Computed Tomography



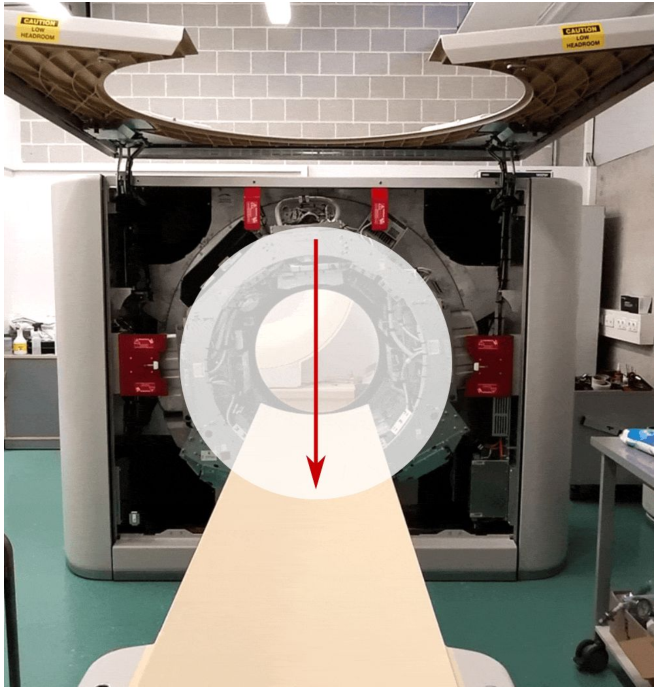
Willer et al., Lancet Digital Health, 2021

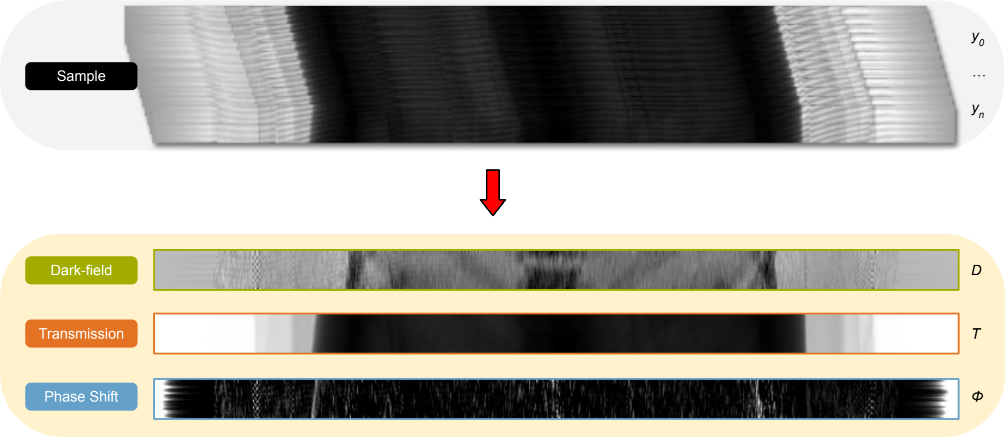
The Physics: Signal Formation



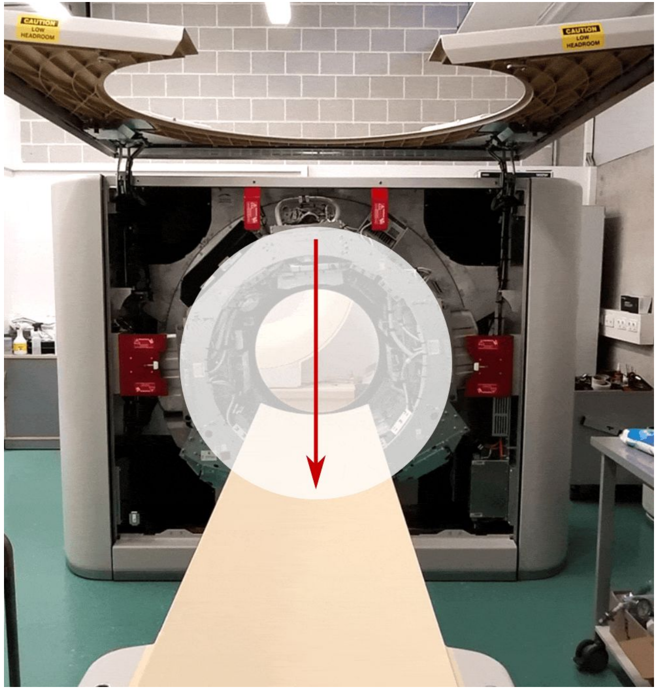


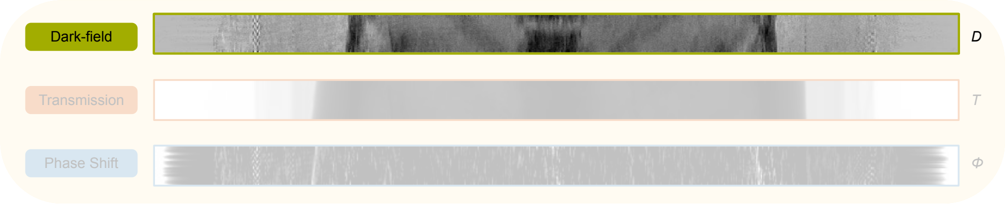
The (Inverse) Problem: Phase Retrieval

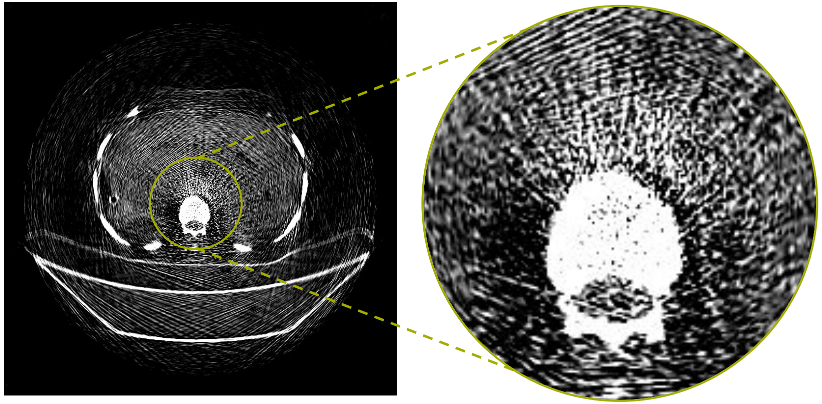




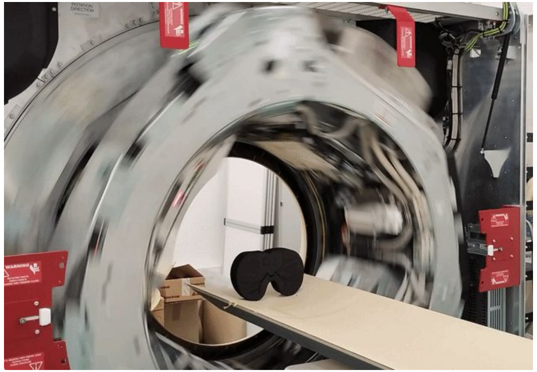
The (Inverse) Problem: Reconstruction Artifacts

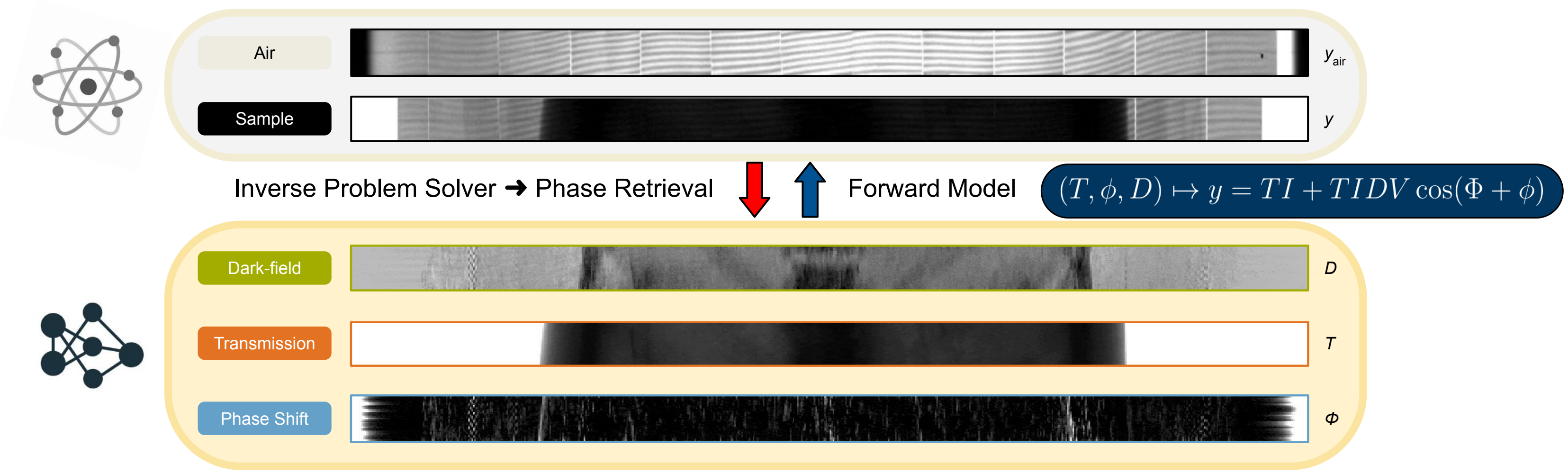






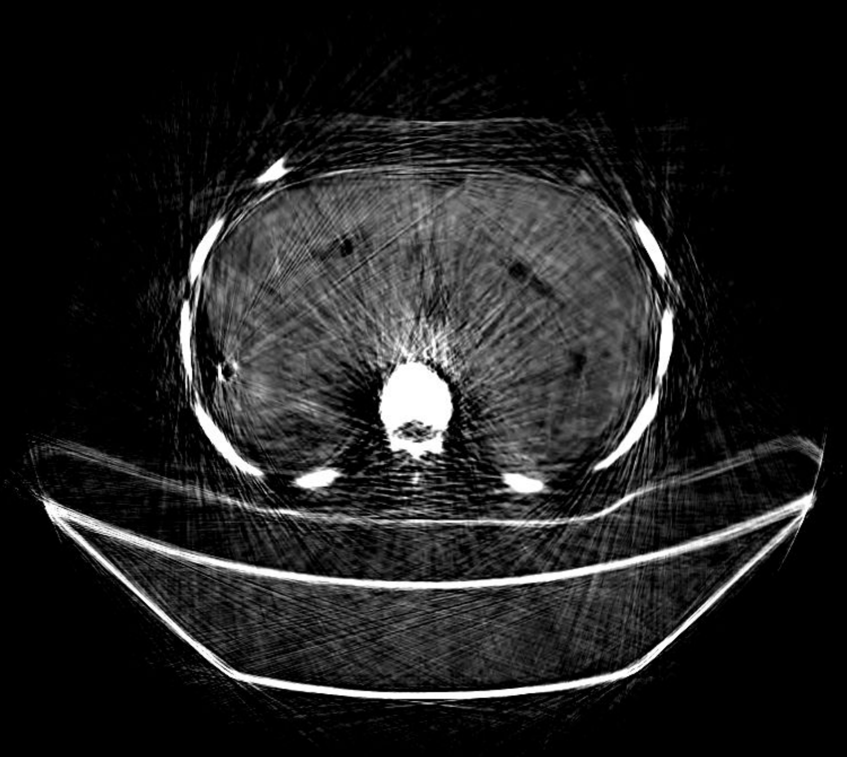
The Solution: A Physics-constrained Neural Field

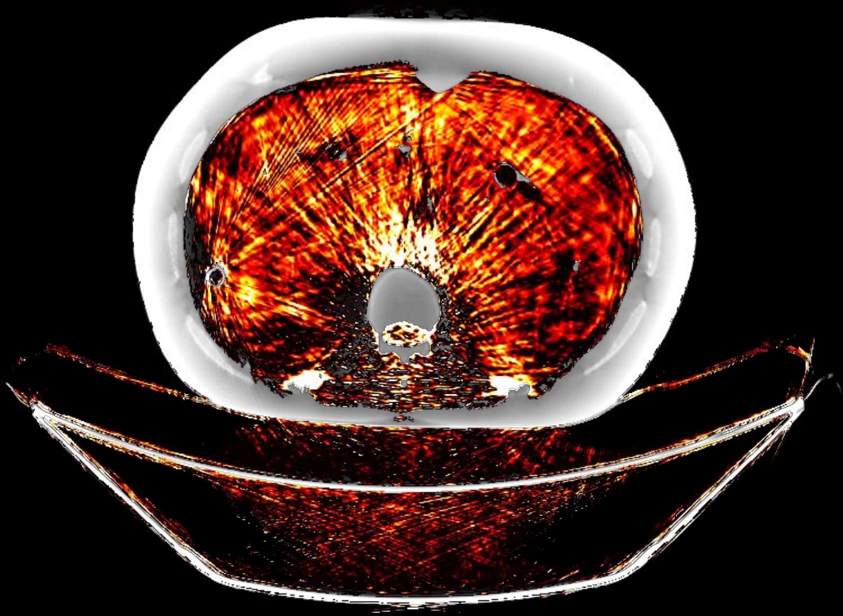




The Neural Field

The Reconstruction





Thank you!

Project webpage:
dcfrey.github.io/sonar

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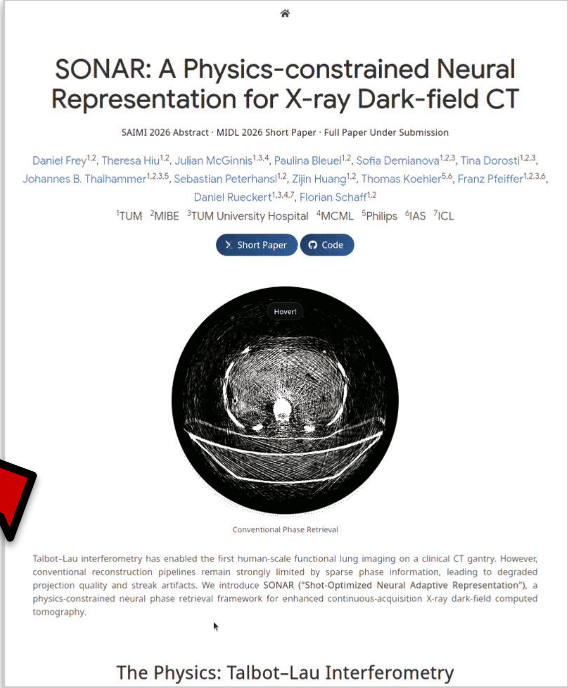
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SAIMI 2026 Abstract · MIDL 2026 Short Paper · Full Paper Under Submission

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Short Paper | Code

Conventional Phase Retrieval

Talbot-Lau interferometry has enabled the first human-scale functional lung imaging on a clinical CT gantry. However, conventional reconstruction pipelines remain strongly limited by sparse phase information, leading to degraded projection quality and streak artifacts. We introduce SONAR ("Shot-Optimized Neural Adaptive Representation"), a physics-constrained neural phase retrieval framework for enhanced continuous-acquisition X-ray dark-field computed tomography.

The Physics: Talbot–Lau Interferometry

Technical University of Munich

Bayerisches Staatsministerium für Wissenschaft und Kunst

Federal Ministry of Education and Research

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